

**BREAST CANCER: EVIDENCE FOR A RELATION
TO FISSION PRODUCTS IN THE DIET**

Ernest J. Sternglass and Jay M. Gould

To establish the possible relation between breast cancer mortality and low doses of radiation due to fission products in the environment, the mortality rates in the nine census regions of the United States for the years 1984–1988 were correlated with the cumulative airborne releases from all the nuclear plants in each region for the period 1970–1987. A high correlation coefficient of 0.91 was obtained for a logarithmic dependence on the total releases, consistent with an indirect action via free-radical oxygen at very low dose rates, in contrast to a direct action on DNA at high dose rates, explaining the wide differences in risk per unit dose obtained in earlier studies. The recent temporal changes of breast cancer rates in the New York metropolitan area including nearby Connecticut, Westchester, and Long Island were examined in relation to the releases from nearby nuclear plants and found to be consistent with a dominant role of short-lived fission products in drinking water and fresh milk. The results support a major role for nuclear plant releases in industrial countries in the recent rises of breast and other forms of cancers not related to smoking, especially among older persons, and strongly support the need to replace nuclear reactors with more benign ways to generate electricity.

Within the last few years, a series of studies have documented a sharp rise in the incidence and mortality rates for most types of cancer not related to smoking, particularly among older age groups, in the industrial countries of Europe, North America, and Asia (1, 2). In the case of breast cancer, there has been a particularly high rate of increase in the incidence among both younger and older women since the mid-1970s in the northeast region of the United States (3), such as Long Island, New York, and Connecticut, that has caused widespread public concern. Although environmental factors of some form, such as pesticides and radioactive fission products in the environment and workplace, have long been suspected to play a role (2, 4, 5), it has been difficult to prove a causal connection in the general population. This is due to the difficulty of obtaining quantitative data on exposure levels to carcinogens that are adequate to clearly establish a dose-response relation

down to the very small trace amounts of these contaminants actually encountered in the diet and drinking water.

It is the purpose of this article to show that for the case of at least one class of known carcinogens—namely, fission products released from commercial nuclear power plants—there is a simple way to establish the existence of a dose-response relation between levels in the environment and cancer mortality in the general population with sufficient precision to determine its form, and thereby to understand the basic nature of the dominant type of biological mechanism involved at very low doses. Moreover, we will show that for the case of breast cancer in the New York metropolitan area, it is possible to identify the principal pathways through which the fission products reach the human body, so that public health measures can be instituted to reverse the recent rises in incidence and mortality rates.

METHODOLOGY

In order to test whether a quantitative dose-response relation exists between exposure to fission products in the environment and breast cancer mortality, we decided, for several reasons, to examine the data for the United States by census regions, for which reliable information on both radioactive releases and mortality is available. First, regional data involving a group of states provide the larger populations needed for strong statistical power, compared with the few sparsely populated counties near nuclear plants that were examined in most earlier studies (6). Second, it is possible to get uniformly collected and officially published data on releases of fission products for each region over a long period of time because the Nuclear Regulatory Commission (NRC) requires detailed annual reports of releases for each commercial nuclear reactor in the United States, and these reports are publicly available (7). Third, although individual doses received will vary depending on dietary habits, sources of drinking water, and distance from the reactor, the relative values of the dose to the average person in each region will be proportional to the total amounts released per capita. This is because the group of states in each region is so large that most of the drinking water, milk, meat, and vegetables consumed in that region will have originated in that group of states, and a comparable fraction of the releases will end up in the human diet.

RESULTS

Figure 1 shows the annual and cumulative releases of airborne fission products in the form of particulates other than rare gases, consisting of iodine-131, strontium-90, strontium-89, cesium-137, and many other radioactive chemicals. Only those isotopes with half-lives greater than eight days are reported, but additional isotopes with half-lives of a few hours to a few days are also released

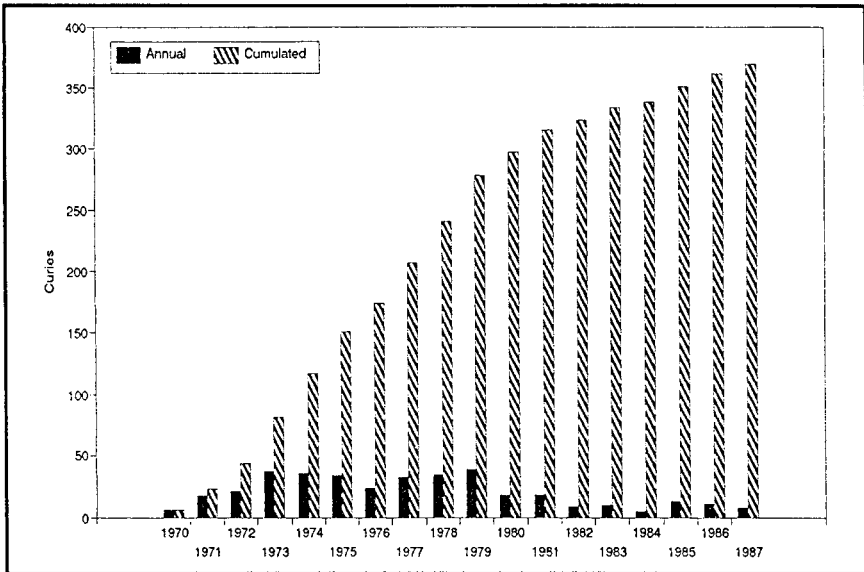


Figure 1. Annual and cumulative airborne ^{131}I and other fission products in the form of particulates released from U.S. nuclear power plants, 1970–1987 (7).

that can be significant when rapid transport of air, drinking water, milk, and fresh vegetables takes place.

Inspection of Figure 1 shows that there were very large releases in the early to mid-1970s when the new 1,000-megawatt plants began to operate, leading to a rapid rise in the cumulative amounts discharged to a total of 370 curies by 1987. Following the accident at Three Mile Island near Harrisburg, Pennsylvania, in 1979 in which an estimated total of 14.2 curies was discharged, improved hold-up and filtration systems led to a general reduction of the airborne releases. A number of individual plants, however, had serious fuel-failure and waste-treatment system problems that in some cases led to annual releases almost as large as those from Three Mile Island.

These amounts are biologically very significant, considering that one curie is a trillion or 1012 picocuries, the units in which fission products in the milk and general diet are measured and regarded to be hazardous. According to Archer (8), who found a strong correlation between rises in leukemia and ^{90}Sr across the United States during the period of atmospheric bomb tests, the diet contained between 6 and 26 picocuries per kilogram of ^{90}Sr in 1964–1965, when peak values occurred in the various regions of the country following the second series of atmospheric tests of 1961–1962; and the mean concentration in cow's milk at the end of the first series ranged from 4 to 16 picocuries per liter in 1959–1960. The total dose produced from the ingestion of 10 picocuries per day of ^{90}Sr by a child in the course of a single year

(365 days) may be estimated, with the dose factor used by the NRC (9) to evaluate the doses from nuclear reactor releases of 1.70×10^{-2} millirads per picocurie, at 62 millirads. This is of the same order as the average yearly dose from natural background radiation in the United States from all external and internal sources of gamma and cosmic rays, estimated to be about 87 millirads (10).

The published data on the airborne releases of "iodine-131 and particulates" and the per capita releases for each of the nine census regions of the United States between 1970 and 1987 are listed in Table 1, together with the average age-adjusted breast cancer mortality rates in each of these regions for the years 1984–1988 [published by the National Cancer Institute (3)] weighted by the number of cancer deaths in each state of the region. These data are also plotted in Figure 2 on both a linear and a logarithmic scale for the releases.

A very close relation is evident between the cumulative releases from 1970 to 1987 and the breast cancer mortality rates. The relation is a concave downward "supralinear" curve (Figure 2, top), or a straight line when plotted against the logarithm of the dose (Figure 2, bottom). The correlation coefficient is 0.91, and the statistical probability that such a high correlation occurred by chance is less than 1 in 1,000 ($P < 0.001$).

Thus the incremental risk per unit radiation dose measured by the steepness of the slope is very high at low doses but levels off to a much lower slope or risk per unit dose at high doses. This explains why short, high dose rate exposures such as those in diagnostic or therapeutic uses of radiation or from the gamma ray flash of a nuclear detonation are so much less serious per unit dose than small, chronic exposures at very low dose rates from inhaled or ingested fission products, as observed for the case of free-radical dominated effects discovered by Petkau (4, 11–14).

The shape of the logarithmic dose-response curve also explains why a linear extrapolation from different population groups exposed to different high doses grossly underestimates the risk at low doses, since the slope of the curve that measures the increase in effect per unit increment in dose increases inversely with the dose received in a fixed time period. Thus, at typical annual dose rates from fission products of 10–100 millirads comparable to those from natural background sources, the slope or incremental risk of a small additional dose is 1,000 times greater than at 10–100 rads, the order of magnitude of the doses received by different groups of A-bomb survivors at different distances from the point of detonation.

CHANGES IN BREAST CANCER MORTALITY IN THE NEW YORK METROPOLITAN AREA

The reactors located most closely to Long Island and New York City are the Indian Point nuclear plant on the Hudson River near Peekskill in Westchester County, 20 miles north of New York City, whose first reactor began operation in 1961; Haddam Neck near Middletown, Connecticut, some 12 miles from

Table 1

Age-adjusted breast cancer mortality rates, 1984–1988,
and iodine-131 and particulates airborne emissions
cumulated from 1970 to 1987, by region

Region ^a	Breast cancer deaths, 1984–88 ^b	Age- adjusted mortality rate, per 100,000 ^c	Standard error	¹³¹ I and particulates, 1970–87, curies ^d	Curies per million persons
New England (NE) ME, NH, VT, MA, RI, CT	13,350	30.7	0.3	42.71	3.46
Middle Atlantic (MA) NY, NJ, PA	39,431	30.6	0.1	129.36	3.52
East North Central (ENC) OH, IN, IL, MI, WI	37,129	28.6	0.1	135.16	3.66
West North Central (WNC) MN, IA, MO, ND, SD, NB, KS	15,245	26.3	0.2	13.84	0.81
South Atlantic (SA) DE, MD, DC, VA, WV, NC, SC, GA, FL	34,719	26.4	0.1	36.04	0.86
East South Central (ESC) KY, TN, AL, MS	11,453	24.7	0.3	2.64	0.18
West South Central (WSC) AR, LA, OK, TX	16,756	23.1	0.2	1.10	0.05
Mountain (MT) MT, ID, WY, CO, NM, AZ, UT, NV	6,269	24.4	0.4	0.35	0.03
Pacific (PAC) WA, OR, CA	26,132	27.1	0.2	8.74	0.29
Continental U.S.	200,484	27.3	0.1	369.94	1.64

^aStates are denoted by postal code abbreviations.

^bSource: The number of breast cancer deaths and age-adjusted mortality rates, from 1984 to 1988, are given for each contiguous state in reference 3, Table I-30.

^cRegional mortality rates are calculated by weighting each state rate by the number of deaths in that state. The age adjustment is to the 1970 U.S. standard population.

^dSource: reference 7, Tables 4 and 5.

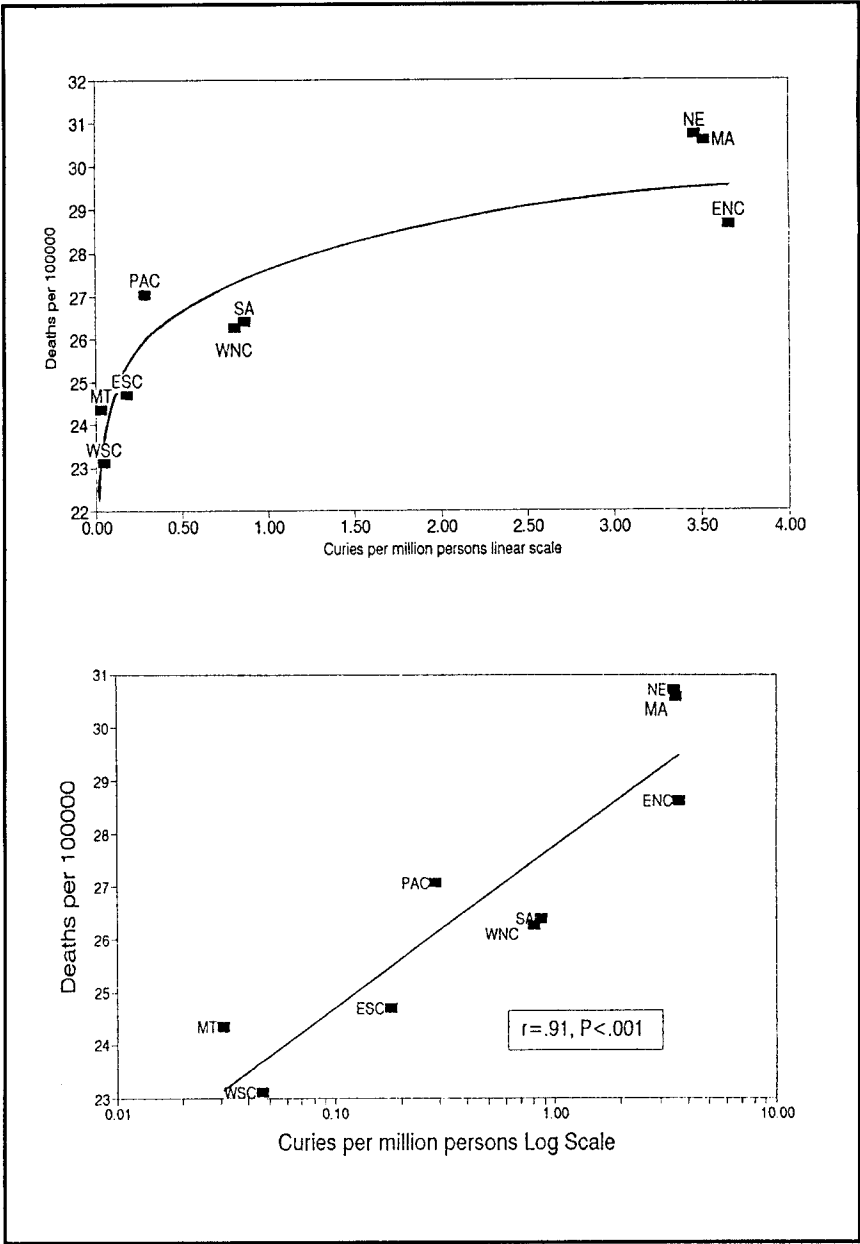


Figure 2. Breast cancer mortality rates by regions of the United States (see Table 1 for explanation of region abbreviations), 1984–1988, versus the cumulative airborne amounts of ¹³¹I and other fission products in particulate form released by commercial nuclear power plants for the years 1970–1987 in each region.

Long Island Sound along the Connecticut River, which started in 1967; and the Millstone plant in Waterford, Connecticut, near New London, located on the coast only about ten miles from Suffolk County at the northern tip of Long Island, which started in 1970.

An examination of their combined releases shows that the amount of ^{131}I and “particulates,” which include other fission products such as ^{89}Sr and ^{90}Sr , discharged into the atmosphere from the stacks of these plants between 1970 and 1987 was more than three times the 14.2 curies reported for the Three Mile Island accident in 1979 (7).

Of all the nearby reactors, Unit 1 at the Millstone plant that went into operation in 1970 had the largest reported releases. Figure 3 shows the reported airborne and liquid releases of fission products from the Millstone Unit 1 reactor for the period 1970–1987. This reactor not only is the closest to Long Island but also reported the highest emissions of airborne and liquid fission products ever released from any nuclear reactor in the United States, much larger than from the Haddam Neck or Indian Point plants. The liquid releases of fission and neutron-activation products into Long Island Sound were close to 200 curies in both 1974 and 1975, while the airborne stack releases of ^{131}I and particulates were close to 10 curies in 1975, and for the years 1974–1978 combined were nearly 24 curies, compared with 14.2 curies reported as having been released during the Three Mile Island accident.

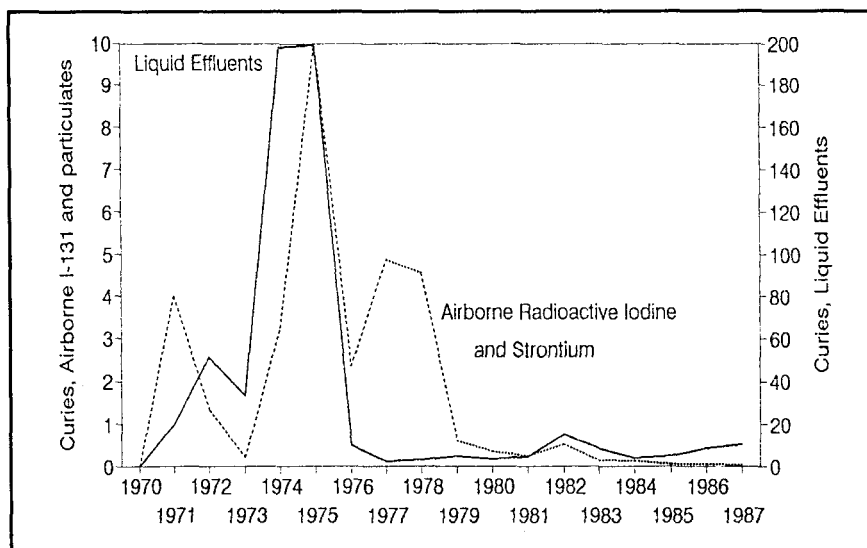


Figure 3. Airborne releases of ^{131}I and other fission products in the form of particulates and liquid releases of mixed fission and activation products into Long Island Sound from the Millstone nuclear plant near New London, Connecticut, 1970–1987 (7).

Altogether, the Millstone Unit 1 reactor released some 32.6 curies of airborne ^{131}I and other fission products, and 581 curies of liquid fission products by 1987, most of it in the period 1972–1979. This represents 7.6 million picocuries per person in Connecticut, Westchester, Suffolk, and Nassau, compared with 3.46 million picocuries per person in all of New England, the highest in the nation. Since one curie is a trillion picocuries, the unit in which the radioactivity of ^{90}Sr and ^{131}I in milk and water was measured and regarded to be of public health concern during the height of nuclear testing, it is clear that these discharges represent a very large contamination of the air, the water, and the diet of the area, comparable to that which occurred during the height of atmospheric testing.

This is brought out by examining the levels of ^{90}Sr in the milk at different distances from the Millstone plant as measured by that utility and by the EPA in the summer of 1976, following the year of the largest airborne releases, and comparing them with those recorded during the height of nuclear testing (Figure 4) (15). Strontium-90 levels in the milk at the closest sampling station 5 miles northwest of the plant reached a concentration of 26 picocuries per liter,

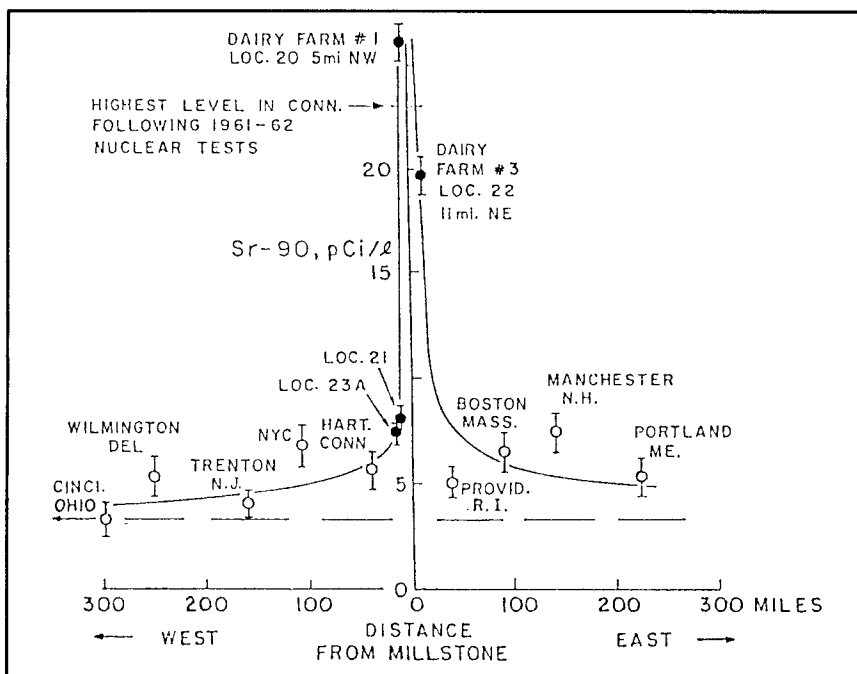


Figure 4. Strontium-90 levels (in picocuries per liter) measured in milk at various distances from the Millstone nuclear plant near New London, Connecticut, July 1976 (15). Sources: N.E. Utilities Environmental Report, 1976, Table 7; and EPA Report no. 7, January 1977, Table 7.

higher than the 23 picocuries per liter yearly average reported for Connecticut in 1961–1962.

Moreover, levels above the lowest in the eastern United States of 3 picocuries per liter were reported as far away as 230 miles in Portland, Maine, downwind to the northeast and to the southwest as far away as New York City and Wilmington, Delaware, decreasing with distance as expected for releases from a stack. Thus, there can be no doubt that these enormous releases reached nearby Suffolk and Nassau counties in Long Island as well as Westchester north of New York City.

Figure 5 shows the incidence of breast cancer in Connecticut for women in the 50–74 age group per 100,000 total population from 1935 to 1990 (16). After being nearly constant or even declining slightly during the period 1935–1944, before any nuclear fission products existed, breast cancer incidence began to rise at a nearly constant rate until 1970, shortly after Haddam Neck and Millstone began to operate, after which the rate of rise increased sharply. The first peak occurred in 1974–1975, some seven to nine years after the first releases from Haddam Neck in 1967, while a second broader rise above the 1945–1970 trend line occurred in 1983–1985, some seven to nine years after the largest releases from the Millstone Unit 1 reactor in 1974–1976.

Such a relatively short delay in the rise of breast cancer and a number of other types of cancer was also seen in Japan following the arrival of fallout from the first nuclear bomb detonations (17, 18). It is also consistent with a relatively early rise in incidence of another solid tumor, thyroid cancer, observed in Belarus only four to five years following the arrival of the Chernobyl fallout (19), presumably because the wide mixture of different fission products acts as both an initiator and a promoter.

The same pattern of sharply rising breast cancer rates is seen in the two counties on Long Island just across from the two Connecticut nuclear plants (20). As can be seen in Figure 6, broad peaks occurred in the rises of breast cancer mortality in Nassau and Suffolk Counties that parallel the peaks of incidence in Connecticut. Thus, there is a peak in both Nassau and Suffolk Counties in 1974–1976, seven to nine years after start-up of the Haddam Neck reactor in 1967, and in 1983–1985, seven to nine years after the largest releases from the Millstone reactor in 1974–1976. Moreover, the greatest percentage rise in mortality rates between 1970–1972 and 1987–1989—39 percent—took place in Suffolk, closest to the Millstone plant near New London, which is only some 10 miles across Long Island Sound from this plant and still the most rural in character. On the other hand, in Nassau, a much more densely populated suburban area to the west at a much greater distance from the Millstone plant, breast cancer mortality rose only 20 percent, and in New York City at the western end of Long Island more than 60 miles away, it actually declined 6 percent during the same period.

However, it is interesting to note that the rise in Nassau took place some 10 years earlier than that in Suffolk, consistent with the earlier start (1961) of the Indian Point Unit 1, which is located only about 25 miles upwind to the northwest

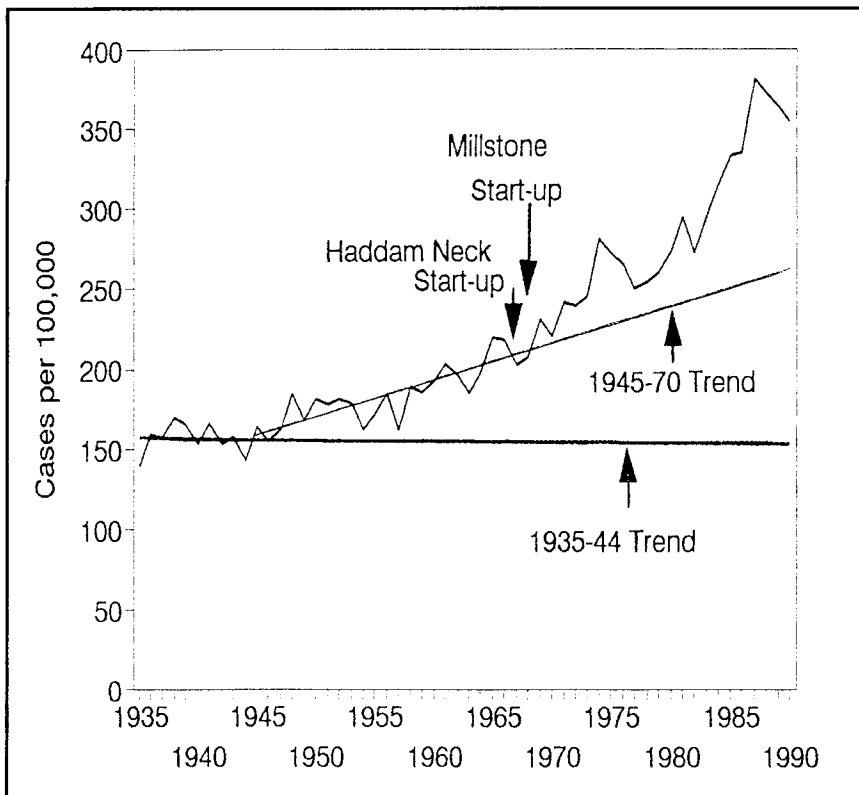


Figure 5. Yearly incidence rates of female breast cancer in Connecticut for the 50–74 age group, 1935–1990. Trend lines shown are for the two periods 1935–1944, before any nuclear fission releases occurred, and 1945–1970.

from the Nassau County border, so that Nassau reached the same rate as Westchester by 1975–1976, exceeding that of New York City thereafter.

CHANGES IN BREAST CANCER MORTALITY IN WESTCHESTER, NEW YORK CITY, AND THE REST OF NEW YORK STATE

Turning next to the breast cancer mortality changes in New York City and Westchester since 1960 (Figure 7), it is evident that before 1976, New York City and Westchester had closely similar rates, except for the years 1964–1967, after the first of three reactors that began to operate at Indian Point in 1961 experienced serious fuel failures accompanied by large emissions (21), which apparently accelerated the spread of existing tumors. With the replacement of the defective

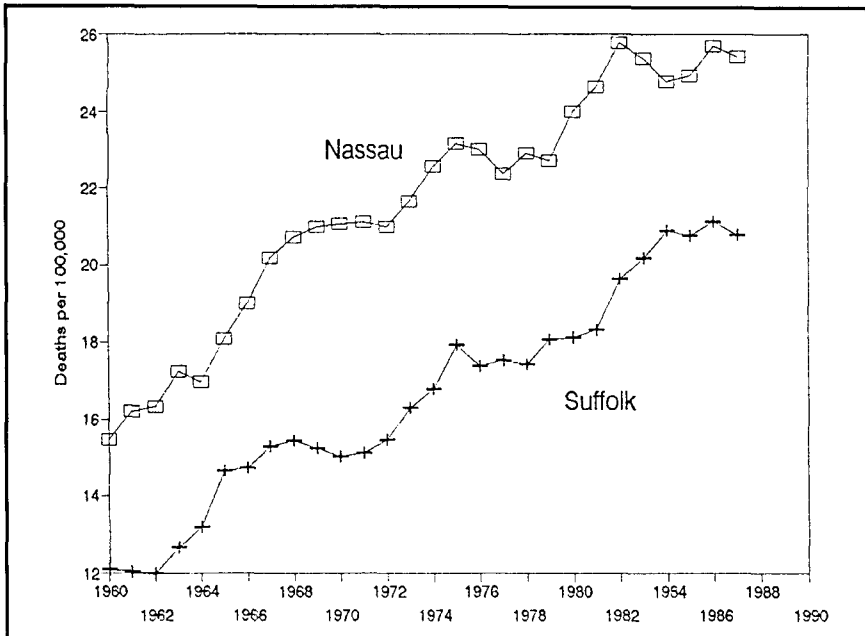


Figure 6. Breast cancer mortality rates in Suffolk and Nassau Counties, Long Island, for the period 1960–1987 (three-year moving average). Note especially the peaks for 1973–1976 and 1982–1987, similar to those in the Connecticut incidence rates.

fuel in 1967, the cancer rate gradually fell so that by 1975 it reached a low point, 5 percent less than that of New York City. However, shortly after Unit 2 went into operation in 1973 and record high releases took place from Units 1 and 2 in 1975, followed by additional releases from Unit 3 going into operation in 1976 (7), the breast cancer mortality rate started to rise sharply once more, reaching an all-time high in 1985 that was 29 percent above the rate for New York City (20).

From 1960 to 1976, the New York City mortality rate also rose in parallel with that of Westchester, following the combined rise of environmental fission products from the large 1961–1962 atmospheric bomb-test series and the releases from the first two Indian Point reactors. However, after 1976 the New York City rate began to decline, except for a small rise in 1980, until it reached a low point 9 percent smaller than the 1980 peak in 1985 and 29 percent below that in Westchester, when another series of large releases occurred from Unit 2 (7).

The most surprising aspect of these remarkable changes in breast cancer mortality rates is that New York City did not continue to rise in parallel with Westchester after 1975, but instead declined throughout the early 1980s. This is contrary to the pattern in all the immediate suburban areas, whose populations

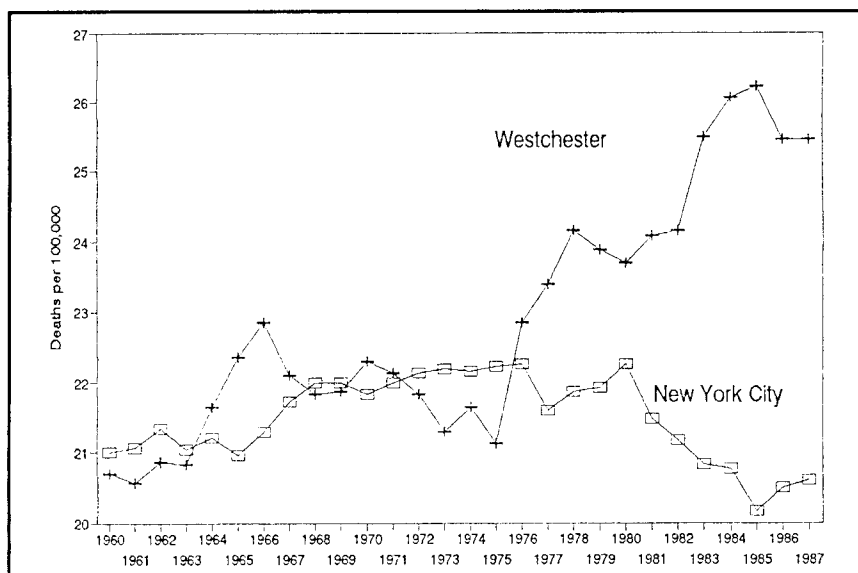


Figure 7. Breast cancer mortality rates in Westchester County, New York, and New York City for the period 1960–1987 (three-year moving average). Note the sharp divergence in trends beginning in the mid-70s, following the completion of the new water supply system for New York City in 1967.

have much higher socioeconomic and educational levels and thus greater access to early diagnosis and therapy than the poorer population of the inner city. Clearly, any increase in the use of mammography during the late 1970s and early 1980s should have had at least as great a downward effect on mortality rates in Westchester, Nassau, and Suffolk as in New York City. Instead, the rates rose in the wealthy suburbs and declined in the inner city, a pattern for which another explanation is evidently needed.

Since neither New York City nor Westchester produce their own milk, meat, and vegetables, it appears that the answer may lie in the changes that took place in the drinking water supply for these two areas in the 1960s (22). In 1837 New York City began to create a series of reservoirs in the Croton River watershed located in Westchester near Peekskill, and started to derive its drinking water from this area. But as the city grew and the Croton area became more densely populated, the demand for more and cleaner water led to the development of two new sources of water in the Catskill Mountains and the headwaters of the Delaware River, some 50 to 150 miles to the west. Large pipelines were built to bring the water into New York City, and by 1967 the gigantic new water supply system was completed. Thus, under normal conditions, about 90 percent of New York City's water is now

derived from the new distant reservoirs, and only about 10 percent is still obtained from the Croton watershed, whose edge is located just 5 miles downwind to the northeast from the site of the Indian Point nuclear plant that began to release large airborne emissions of both short- and long-lived fission products in the early 1960s.

Since the Croton reservoirs continued to supply Westchester County with drinking water as more reactors began to operate at the Indian Point site, it is clear that after five to ten years, the cancer rates for Westchester and New York City would be expected to diverge in exactly the directions observed: upward for Westchester and downward for New York City.

In the light of this hypothesis for the changes in the Westchester and New York City breast cancer rates, it becomes possible to resolve still another apparent paradox: the strange divergence and rejoining of the rates for New York City and the rest of New York State between 1947 and 1984. As Figure 8 shows, New York City had a lower mortality rate than the rest of New York State in the period just before 1945 and actually showed a slight decline during this period, as did the Connecticut incidence rates. Then in 1944, fission products were first released

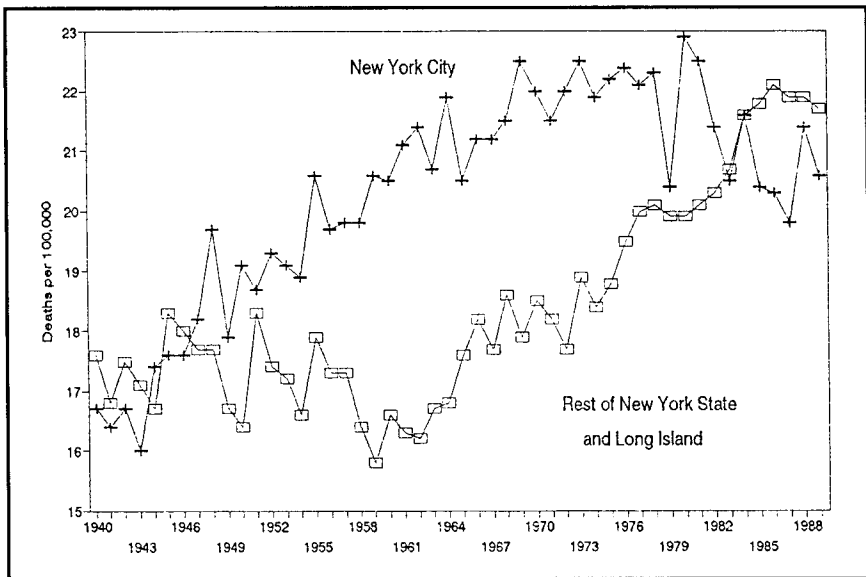


Figure 8. Breast cancer mortality rates in New York City and the rest of New York State including Long Island, 1940–1989. Note (a) the rapid early rise in New York City, which uses surface water, following the start of short-lived fission product releases into the atmosphere in 1944–1945, and (b) the later rise for the rest of the state beginning with the massive increase of long-lived ^{90}Sr in the milk produced by the largest H-tests and the completion of the first of a series of commercial nuclear reactors in the early 1960s.

into the environment on a large scale, beginning with the enormous emissions of ^{131}I and other short-lived isotopes that drifted across the country from the Hanford plutonium production plant beginning in 1944, and the first bomb detonations in 1945 (21).

Within a few years of further releases of these isotopes and as a result of the onset of low-altitude atmospheric testing in Nevada, the Pacific, and Siberia, the New York City breast cancer mortality rate started to rise above that for the rest of the state, while the area outside the city actually kept declining slightly, reaching an all-time low in 1959 despite a series of peaks occurring after known major rain-out episodes. This pattern is consistent with the fact that New York City drinking water comes from surface sources that reach the consumer in a matter of days or weeks, whereas most rural areas, particularly Suffolk and Nassau in Long Island, obtain their drinking water from wells tapping into deep aquifers reached by rain only in a matter of five to ten years (23), sufficiently long to allow the most active short-lived isotopes such as ^{131}I and ^{89}Sr (with half-lives of 8 and 50 days, respectively) to decay. Thus, only when the long-lived ^{90}Sr (half-life 28 years), mainly produced in large quantities only after the first H-bomb tests in 1953, reached the underground water did the rural water supplies become contaminated, explaining the delay in the breast cancer rise in areas outside New York City.

Moreover, beginning in the 1960s when the first nuclear reactors started to operate throughout New York State as well as in nearby New Jersey and Connecticut—namely Indian Point in Westchester (1961), Haddam Neck near Middletown in Connecticut (1967), Oyster Creek near Atlantic City in New Jersey (1969), Nine Mile Point near Syracuse and Ginna near Rochester in Upstate New York (1969), and Millstone near New London (1970)—short-lived activity started to reach the diet everywhere at a growing rate, and the cancer mortality rate for the rest of New York State began to climb to the same level as in New York City.

By 1983–1984, the breast cancer mortality rate in the largely rural areas outside New York City that were using increasingly contaminated well-water began to exceed that for New York City, which was decreasing after the water supplies in the Catskills and Delaware system, located upwind and more than 60 miles from any reactors, began to supply less radioactively contaminated drinking water to the city. The contamination had declined since atmospheric bomb-testing had ended with the last test by China in 1980, and the short-lived fallout from the Three Mile Island accident in 1979 that had reached the Catskill mountain reservoirs had decayed.

Thus, although radioactive contamination of the milk, dairy products, meat, and vegetables continues to be a problem for every region containing nuclear reactors, the source of the water apparently plays a major role in explaining the changes in cancer rates for populations served by different types of water supplies, as in New York City and the nearby suburbs.

DISCUSSION

The results described above are consistent with the recent large-scale prospective studies that do not support a causal relation for any dietary fats in the risk of breast cancer (24). At the same time, the present results support the earlier finding of many studies that the consumption of fresh milk and dairy products is related to a higher risk of breast cancer (25–27), since milk, like surface water, reaches the consumer within a few days after the deposition of short-lived isotopes such as those of iodine and strontium, which concentrate in the nonfat part of both cow and human breast milk.

The hypothesis that fresh fission products released from commercial nuclear reactors play a dominant role in breast cancer rates also explains why Japan continues to have relatively low rates of breast cancer that are increasing only slowly (28), even though many reactors are in operation. Most Japanese women still do not drink milk or consume significant amounts of other dairy products and red meat. The principal source of animal protein is still ocean fish, which contain about 1,000 times less ^{90}Sr per gram of calcium than do freshwater fish (29). In addition, the Japanese diet relies heavily on rice and sea vegetables low in strontium and high in nonradioactive iodine that protects against the uptake of the radioactive form (^{131}I), which concentrates in breast milk and the thyroid. Finally, Japanese reactors are located along the sea shore, so that most of the radioactivity is discharged into the oceans rather than into the air. Drinking water is thus relatively protected, and most fish, such as tuna and salmon, are caught far from the coast of Japan.

The evidence linking nuclear reactor releases to the recent rise in breast cancer rates is independently supported by the fact that internationally, only four of 16 industrial countries studied—New Zealand, Australia, Hong Kong (28), and Israel (30)—showed declines in breast cancer mortality between 1971 and 1986, while rates in all others rose. Of the four industrial countries that showed declines, none had large commercial nuclear reactors operating in or near their borders during this period, while all the others did, consistent with the present hypothesis. (Sweden is no exception, since an apparent recent decline reported by Kohlmeier and associates (28) was subsequently found to have been due to a coding change, as described in the discussion of their paper.) The probability of this pattern being a purely chance result is less than 1 in 1,000, so the hypothesis presented here is strongly supported by the international pattern of recent breast cancer mortality changes.

No major reduction in the use of pesticides has been reported for any of these countries during the 1970s and early 1980s, except in Israel (30). However, the rather large reduction of 34 percent in breast cancer mortality for younger women in Israel since bomb-test fallout ended is consistent with a possible synergistic interaction between pesticides and fission products such as ^{89}Sr and ^{90}Sr , since

Israel had one of the highest rates of breast cancer mortality when worldwide fallout was greatest (30).

Because strontium is chemically similar to calcium, it is primarily stored in bone, where it irradiates the bone marrow in which the cells of the immune system originate. Since the newborn infant requires large amounts of calcium to build its bones during early development, human breast milk, like cows' milk, contains high concentrations of calcium and thus also strontium and other fission products that are stored in the bone. The strontium can be released from these maternal stores when needed to build the bone of the developing fetus and infant. Moreover, in addition to ^{90}Sr , radioactive ^{131}I and other short-lived iodine isotopes are found to be concentrated in both cows' and human breast milk, because iodine is needed for the production of a key hormone required for the normal growth of the developing infant.

Thus, the breast tissue of young women is especially strongly exposed to radioactive fission products, along with the bone and bone marrow where the cells of the immune system originate. Thus the immune defenses against the spread of cancer cells are impaired at the same time as the breast tissue is irradiated by the intense and highly energetic beta particles emitted by these isotopes. The isotopes of strontium are therefore not only initiators but also promoters of tumors, explaining why breast cancer mortality rates begin to increase within only a few years after fresh fission products enter the diet.

The reason that these synthetic bone-seeking isotopes are so much more toxic than radium or radon, which are part of the natural background, is the following. All the radioactive strontium isotopes emit very energetic, low-mass beta particles or electrons that can travel many millimeters in tissue and thus reach the bone marrow. Radium, on the other hand, which also concentrates in bone, emits heavy alpha particles of very short range, so that most of them are unable to reach the marrow. Potassium-40, which accounts for about a third of the total background other than radon in certain homes, is uniformly distributed throughout the soft tissue and body fat, like ^{137}Cs , and thus does not concentrate in bone. Finally, external gamma rays, either from the ground or other energetic sources such as the extremely powerful cosmic rays that produce another third of all background doses, do not concentrate chemically in specific organs or the bone marrow.

The same is true of medical X-rays, which again do not concentrate in any particular organs and do not stay in the body when the X-rays are turned off, unlike strontium, which remains in bone for long periods of time. This leads to chronic exposures at extremely low dose rates, millions of times lower than encountered for medical X-rays. At these very low dose rates, free-radical oxygen molecules are much more efficient at producing biological damage to cell membranes than direct hits on the nuclear DNA, which is the dominant mechanism involved in short, high dose rate exposures such as occur in chest X-rays or mammography (12–14).

This helps to explain why small doses from ingested fission products are 1,000 times more biologically serious in producing breast cancer in postmenopausal women than mammograms, which require doses that are comparable to the annual dose from natural sources per image. Because DNA damage is effectively repaired in a healthy adult, the dose required to double the risk of breast cancer from short X-ray exposures in women over 50 years of age is of the order of 100,000 millirads (3), the dose that would be produced by some 10,000 modern chest X-rays or 1,000 mammograms with today's technology. To express this in another way, the increase in risk for X-ray exposures is about 0.001 percent per millirad, compared with 1 percent per millirad for chronic internal exposures to fission products.

The dominant role of free-radical oxygen at the low dose rates encountered with fission products, which leads to the logarithmic or concave downward or "super-linear" form of the dose-response curve, also explains the recent evidence for a protective role of vitamin C (31), known to be a free-radical scavenger.

CONCLUSION

The evidence for a logarithmic dependence of cancer risk on dose removes the major obstacle to the acceptance by the scientific and public health community of the results of the two dozen studies showing a detectable health impact of extremely small doses due to releases from nuclear facilities (6). Until now—in the absence of a dose-response curve in humans down to the very low doses encountered for the daily, routine releases from commercial nuclear reactors—theoretical estimates of risks have had to be based on a purely speculative extrapolation from high-dose exposures that did not foresee the dominant role of free-radicals and their action on the immune defenses of the body at low dose rates.

The results presented here indicate that the risk estimates for chronic, low-dose exposures derived by linear extrapolation from high doses widely accepted as conservative (4, 5) underestimate the actual risk by 100 to 1,000 times. At the same time, when the logarithmic form of the dose-response curve is taken into account, a single dose-response curve fits (a) the low risk per rad at high doses as deduced from the A-bomb survivor and medical exposure data (4, 5), (b) the higher risk per rad for those occupationally exposed (32, 33), and (c) the still higher risk per rad for the general population that receives chronic exposures at the lowest dose rates (34), as shown by the present study. But when only a few sparsely populated counties near nuclear facilities are examined, as in previous studies, the inherent fluctuations in the small number of cases and the small differences in doses make the effect difficult to detect. It was seen, however, at statistically significant levels in two-thirds of over 40 such studies reviewed by Shleien and associates (6), and in a very carefully planned and executed study by

the Bureau of Health in the State of Maine around the reactor located in Lincoln County, using cancer incidence data (35).

The present study, based on much larger population groups and uniformly reported releases from 90 nuclear plants over nearly 20 years of operations, explains the unexpectedly large risk of the small doses. It thus strongly supports the many smaller studies indicating that radioactive releases from nuclear plants appear to be an important and hitherto neglected factor in the otherwise unexplained recent rise of cancer in most industrial nations. In effect, the results presented here represent the outcome of an enormous, unplanned, double-blind epidemiological study in the United States involving some 200 million human beings exposed to fission products over a dose range of 100 to one for nearly a whole generation.

The effect is found to be greatest in those industrial nations that operate large nuclear reactors near their metropolitan areas, especially when there are nearby surface drinking-water supplies, or when the reactors are located close to the dairy farming areas that supply the cities with fresh milk. This finding is consistent with the 100 to 1,000 times greater concentration of the fission-produced ^{137}Cs relative to naturally occurring ^{40}K in the trees of the United States and Europe, compared with those in Australia and New Zealand, recently reported by Burrows and Chalmers (36) as evidence for the potential importance of low levels of ionizing radiation as a cause of cancer.

The radiation produced by fission products appears to act synergistically with other environmental agents known to be carcinogenic, such as ordinary air pollutants, diesel fumes, dust, asbestos, cigarette smoke, pesticides, herbicides, and exogenous hormones (37). This has been demonstrated in numerous laboratory studies and for individuals exposed occupationally, particularly in the case of uranium miners. Thus, since many other types of cancers such as colon, kidney, bladder, and prostate cancer have also risen more rapidly in parallel with breast cancer in recent years, especially for older persons, in industrial nations using nuclear power (28), and since they show similar recent rises in the United States (3, 38), it now appears that chronic exposure to nuclear fission products in the diet and drinking water may be the single largest factor in the increased incidence of most forms of malignancies in the general population since World War II.

It was the Cold War that drove governments into the hasty construction of reactors and the testing of ever larger number of nuclear weapons under a cloak of secrecy. This secrecy not only resulted in keeping the effects of weapons fallout from the public and the scientific community (30), but also served to hide the possibility of major accidents and large routine releases from military and civilian reactors (40).

Moreover, as revealed in documents made public under the Freedom of Information Act in 1979, and in the law suit won by the cancer victims downwind from the Nevada test site (*Irene Allen et al. vs. The United States of America*, May 10, 1984) (39, 40), a primary concern to keep secret the health effects of radioactive

contamination of the drinking water and diet by the production and testing of nuclear weapons forced the government to downplay environmental contaminants in general as a possible causal factor in the rise of cancer rates. Instead, under pressure from the agencies concerned with the testing of nuclear weapons and the development of nuclear energy, the agencies charged with the protection of public health and environment focused their attention on non-weapons-related effects such as the burning of fossil fuels or the action of radon and ordinary chemicals. Thus cancer research funding was directed into areas other than the effects of low level fission product contaminants in the diet (37).

Since nuclear weapons production and testing were perceived by the military as absolutely essential for national security, as was the operation of nuclear power reactors by the atomic energy agencies of all the major powers, this conviction necessarily led to government policies of concentrating all efforts on finding a cure for cancer rather than its prevention (37). And if the necessary environmental releases were not to be considered significant factors, blaming the lifestyle of the victims for the rising rates of cancer mortality, despite the enormous expenditures on finding a cure, was the only alternative.

Given all the evidence for the serious underestimation of low-dose-rate effects produced by chronic radiation exposures that has come to light in the last 20 years, especially in the case of inhaled or ingested fission products, a prudent concern for public health would seem to require that nuclear reactors be phased out as rapidly as possible. They can be replaced with more benign ways to meet society's needs for energy, such as natural gas and solar, hydro, and wind energy, and conservation measures already used successfully by many nations. With the end of the Cold War, these alternatives to nuclear energy offer a chance to reduce the sharply rising numbers of cancer and other chronic and infectious diseases involving the action of free-radicals, which are driving up the cost of health care, lowering human productivity, and thereby threatening the future economic well-being of the United States and all other nations that continue to operate large nuclear facilities.

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Note added in proof

While preparing a paper for a conference on “Children and Radiation” held at the Norwegian Center for Child Research in Trondheim, Norway, June 10–14, 1993, we found independent support for the relation of breast cancer to fission products in the diet in an analysis of the rise in the percentage of low birth weight infants in the United States that began in 1945 and peaked when ^{90}Sr in adult bone

reached its maximum concentration in 1966. By examining the temporal and geographical patterns of births below 2,500 and 1,500 grams by race, for every state in the United States since 1950 and for New York State outside New York City since 1945, we found that the current wide difference in rates of low birth weight between white and nonwhite infants began only after the onset of nuclear releases. Moreover, with the exception of Nevada where nuclear testing continued until 1992, this wide gap persisted only in the high rainfall states that received the worldwide fallout from the testing of megaton-sized weapons and nuclear accidents, such as those of Three Mile Island and Chernobyl. We also found that the decline in low birth weight after 1966 slowed in the early 1970s, halted, and began to rise once more with the onset of large releases from nuclear power plants, at the same time as the incidence of breast cancer began to accelerate. There is a logarithmic dose-response relation between successive releases from the Indian Point nuclear plant and sharp rises in the percentage of very low birth weight black infants in New York the following years. This is the same form of dose-response relation seen for breast cancer mortality and nuclear plant emissions in the various census regions of the United States.

Direct reprint requests to:

Dr. Ernest J. Sternglass
170 West End Avenue
Apt 27H
New York, NY 10023